

Chapter Three

Sampling and Sampling Distribution

Chapter Outline

- ✓ Sampling theory
 - Basic definitions
 - The need for sampling
 - Types of samples
- ✓ Sampling Distributions
 - Sampling Distribution of the mean.
 - Sampling Distribution of the proportion.
 - Sampling distribution of the difference b/n two means.
 - Sampling distribution of the difference b/n two proportions.

1.1 Sampling Theory

1.1.1 Definition of terms

1. **Population:** Entire number of observations, individuals, objects, items, etc
 - ✓ Involves the totality of the elements being studied.
2. **Sample:** Is part of the population (subset of the population).
 - ✓ Certain portion of the population which believed as the representatives of the population.
3. **Element:** Each individual unit of the population:
 - ✓ Also called building blocks of the population.
3. **Census:** Involves gathering of information from all elements of the population.
 - ✓ Enumeration or measurement of ever individual or item of the population.
5. **Sampling:** - Involves gathering information from some part or elements of the population.
6. **Population parameter (parameter):** - Involves any measurement or characteristics taken from the population. E.g. μ , δ , δ^2
7. **Sample statistics:** - It is the measurement or characteristics taken from the sample. E. g. $\bar{x}$, S , S^2

1.1.2 Why we need Sampling

- To contact the whole population would be time consuming.
- Studying all the items of the population may be costly.
- Physical impossibility of checking all the items of the population. E.g. fish, birds, etc...
- Destructive nature of some tests

- The sample results are adequate.

1.1.3 Types of Sampling/ Sampling Techniques

There are two types of sampling techniques;

1. Random sampling (probability sampling)
2. Non random sampling (non-probability sampling)

Whether a sample is random or non- random sampling, it is based on how it is selected.

1. Random Sampling (Probability Sampling): is a technique in which every unit in the population has a chance (greater than zero) of being selected in the sample.

- With probability sampling, every element of the population has a known probability of being included in the sample.
- In this sampling technique, every individual has an equal opportunity for selection and this can be achieved if the researcher utilizes [randomization](#).
- The advantage of using a random sample is the absence of both systematic and sampling bias. If random selection was done properly, the sample is therefore representative of the entire population.
- Random samples are obtained from a finite or infinite population.
- If we have a finite population “n” and if we are asked to find the number of different samples of size “r” that can be taken from the population, we use the following formula.

$$\text{No. of samples of size } n = C_r^n = \frac{n!}{r!(n-r)!}$$

Where n= number of elements in the population

r = the proposed sample size

Example 1

Suppose there are 8 persons from whom we have to select samples of size 3. How many samples can be selected? Ans: 56

Example 2

A group of 10 students are to be divided in to two groups of 5 each and seated at two tables. How many different ways are there of dividing the 10 students? Ans. 252

- ***There are four types of random sampling techniques. They are:***

- A. Simple random sampling
- B. Systematic random sampling

C. Stratified random sampling

D. Cluster random sampling

A. Simple Random Sampling

- If a sample is selected in such a way that every unit or element in the population has an equal chance /probability being selected, it is simple random sampling.
- It is the simplest method of random sampling techniques
- Each and every element has equal chance of being selected.
- No personal bias
- Events are equally likely to occur

Example: Consider four peoples A, B, C, and D asking for vacation but only two of them can leave at the same time.

$$\text{Possible combination of two people} = C_n^N = \binom{N}{n} = \frac{N!}{n!(N-n)!} = \frac{4!}{2!(4-2)!} = 6$$

AB, AC, AD, BC, BD, CD

A - Three times $\Rightarrow P(A) = 3/6 = 1/2$

B - Three times $\Rightarrow P(B) = 3/6 = 1/2$

C - Three times $\Rightarrow P(C) = 3/6 = 1/2$

D - Three times $\Rightarrow P(D) = 3/6 = 1/2$

B. Systematic Random Sampling

- 📊 It is a method in which a random starting point is selected and then every n^{th} member of the population is selected.

📊 Steps:

- ✓ Arranging / ordering the values of the population.
- ✓ Determine the K^{th} value, and the interval N/n ,

Where N = the population size

n = desired sample size

- ✓ Selecting the starting point by using simple random sampling.

C. Stratified Random Sampling

- The population is divided into sub groups called strata; and a sample is randomly selected from each stratum.
- It is used when the population consists of a number of heterogeneous sub-population groups (stratum).

D. Cluster Random Sampling

- ✓ It is important to reduce the cost of sampling from a scattered population in a large geographic area.
- ✓ Population is divided into clusters using natural occurrence, geographic or other boundaries.
- ✓ The clusters are randomly selected and a sample is collected by randomly selecting from each cluster.

2. Non-Probability(Non-random) Sampling

- ❖ This technique does not allow the random procedure. Here, each element doesn't have an equal chance of being selected. A non random sample is selected by a procedure other than probability considerations.

E.g. a sample may be chosen as per the convenience of the person selecting it. It may be selected by an expert on his judgment.

- ❖ There are three types of non-random sampling.

They are:

- A. Quota sampling
- B. Judgment sampling
- C. Convenience sampling

A. Quota Sampling

Quota sampling involves the fixation of certain quotas. Suppose that in a certain territory we want to conduct a survey of households. Their total number is 200,000. It is required that a sample of 1 percent, which is 2000 household, is to be covered. We may fix certain controls. A sample of this size can be selected subjected to the condition that 1,200 household should be from rural areas and 800 from the urban areas of the territory. Likewise, another quota can be fixed. Of the 2000 sample households, rich households should number 150, the middle class ones 650 and the remaining 1,200 should be from the poor. If you select a sample in such a way by fixing quotas, it is called quota sampling.

B. Judgmental Sampling

The main characteristic of judgmental sampling is that units or elements in the population are purposively selected. It is because of this that Judgment samples are also called purposive samples. When small sample of a few units is to be selected, a judgment sample may be more suitable as the errors of judgment are likely to be less than the random errors of a probability sample. However, when a large sample is to be selected, the elements of bias in the selection could be quite large in the case of a judgmental sample.

C. Convenience Sampling

This method is based on the convenience of the statistician who is going to select the sample. This type of sampling is also called accidental sampling as the respondents in the sample are included in it merely on account of their being available on the spot where the survey is in progress.

1.1.4. Sampling and Non-Sampling Errors

- ✓ A number of samples of the same size can be selected from a population. Different samples selected from the same population will give different results as the elements included in the sample are different. This will give rise to sampling errors. When there is a difference between a sample statistics,

say, samples mean and population mean, then that difference is called error.

- ✓ Sampling error is the difference between a sample statistics and its corresponding population parameter.
- ✓ Since the sample is part/portion of the population, it is unlikely that the sample mean would be exactly equal to the population mean.

Example: consider the output population of five employees: 97, 96, 103, 99, and 105, and take a sample of the output of two employees.

$$\mu = (97 + 96 + 103 + 99 + 105) / 5 = 100$$

$$\bar{x} = (97 + 105) / 2 = 101$$

$$\text{Error} = \mu - \bar{x} = 100 - 101 = -1$$

- ✓ There are two types of errors. They are:

1. Sampling errors
2. Non-sampling errors

Sampling Errors: Sampling errors are errors that occurred due to the imperfect selection of the appropriate sampling technique.

Non- Sampling Errors: Are results from some imperfect of the sampling process. Include all sources of errors other than those introduced by the random sampling procedures.

There are two types of non-sampling errors:

1. Administrative errors
 - Data processing errors
 - Sample selection errors
 - Interview errors
2. Respondent errors
 - Non response
 - Response bias

1.2 Sampling Distribution

- ✓ It is the probability distribution of all possible sample statistics for a given sample size.
- ✓ There are as many sampling distributions as there are many sample statistics. Example:
 - Sampling distribution of the mean.
 - Sampling distribution of the standard deviation
 - Sampling distribution of the range.
 - Sampling distribution of the proportion

1.2.1 Sampling Distribution of the Sample Mean

Sampling distribution of the sample mean is the probability distribution of all Possible sample means of a given sample size.

Example 1

Suppose that each of the four typists making up a population of secretarial support service in a company is asked to type the same page of a manuscript. The number of errors made by each typist is presented as follows.

Typist	Number of errors made
Ann	3
Bob	2
Carla	1
Dave	4

Required:

- Calculate the mean and standard deviations of the population.
- How many different samples of size 2 can be selected from the population?
- Construct sampling distribution of the sample mean for a sample size of two.
- Compute the mean and standard deviations of the distribution.
- Compare the values in A and D and what conclusions can be made.

Example 2: The scores of five students A, B, C, D, E out of six marks is as follows:

A	B	C	D	E
0	3	6	3	5

Required:

- What is the population mean and population standard deviation?
- How many different samples of size two can be selected?
- Construct sampling distribution of the sample mean for a sample size of two?
- What is the mean and standard deviations of the sampling distribution?
- What conclusions can be made about the population and the sampling distribution?

Exercise: Given five elements of a population; 3, 6, 12, 15, 18

Required:

- Compute population mean and standard deviation?
- How many different sample of size 3 are possible?
- Construct sampling distribution of the $\bar{x}$ for sample size of three?
- Compute the mean and standard deviation (standard error) of the sampling distribution of the $\bar{x}$?
- Compare
 - Population mean and mean of sampling distribution of the sample mean?
 - Population standard deviation and standard deviation of the sampling distribution?
- Does the mean of sampling distribution of the sample mean shows some tendency towards

being normal distribution?

❖ *Central Limit Theorem*

It states that:

1. If the population is normal distribution, then sampling distribution of the sample means (distribution of the sample means) is also normal for all sample sizes.
2. The sampling distribution of sample means practically normal if the sample size is $n \geq 30$. Normal sampling distribution of the sample means can be converted in to standard sampling distribution of the sample means by:

$$Z = \frac{\text{Value of RV} - \text{Mean of RV}}{\text{Standard deviation of the RV}}$$

$$Z = \frac{\bar{X} - \mu_x}{\sigma_{\bar{x}}}$$

Where:

$\bar{X}$ = sample mean

$\mu_x = \mu_{\bar{x}}$ = mean of the sample mean = mean of the population

$\sigma_{\bar{x}}$ = standard error of the mean

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$

- According to central limit theorem, for a large sample size, the sampling distribution of the sample means is approximately normal regardless of the shape of the population distribution. The sample size is large if the $n \geq 30$.
- As the sample size increases, the shape of the sampling distribution of the sample means tends to be more and more approximate to normal distribution.

Example 1: Hourly wages of workers in an industry have a mean rate of Br 5 per hr. and standard deviation of 0.60. What is the probability that the mean wage of random sample of 50 workers will be;

- A. Greater than BR 5.10
- B. Less than br.4.80
- C. b/n.4.80 &5.10

Exercise 1: Consider a firm Manufacturing Electronic watches whose useful life is approximately normally distributed with mean life of 5 years and standard deviation of 1.5 year. What is the probability that a random sample of 25 watches has an average life of

- A. Between 5 and 6 years?
- B. Between 4 and 5 years?
- C. Less than 6 years?
- D. Less than 4 year?

Exercise 2: National Examination Bureau (NEB) of Ethiopia has found that the mean score of the 2000's examinees is 250 and the standard deviation of the scores is 48. The scores are normally distributed. If a

random sample of 16 examines is taken,

- A. What is the probability that the sample mean score will be
 - i. Greater than 250?
 - ii. Greater than 270?
 - iii. Less than 230?
 - iv. Between 230 and 270?
- B. What Minimum average score does the highest 5 percent score of the examinees have?
- C. About what two values will
 - i. 68% of the mean scores fall?
 - ii. 95% of the mean scores fall?
 - iii. 99.7% of the means

1.2.2. Sampling Distribution of the Difference b/n two Sample Means

If independent random samples of size n_1 and n_2 are drawn (with out replacement) from two different finite populations

	Population 1	Population 2
Population size	N_1	N_2
Mean	μ_1	μ_2
Standard deviation	σ_1	σ_2
Sample size	n_1	n_2

The mean and standard deviations of the distribution are given by:

Mean	$\mu_{\bar{X}_1} - \mu_{\bar{X}_2} = \mu_{x_1} - \mu_{x_2}$	
Standard deviation of the distribution	$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ 	Sampling with replacement & $n \leq 0.05N$
	$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{(\frac{\sigma_1^2}{n_1})(\frac{N_1 - n_1}{N_1 - 1}) + (\frac{\sigma_2^2}{n_2})(\frac{N_2 - n_2}{N_2 - 1})}$ 	Sampling with out replacement

Example: A teacher has been giving tutorial class for four accounting and four business administration female students for 8 weeks. Test scores out of 10 is as follows.

Accounting				Business Administration			
A	B	C	D	F	G	H	I

6	5	7	9
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7	8	4	10
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The teacher wants to take a random sample of two students from each department.

Required:

- Calculate population mean and standard deviation difference.
- Construct sampling distribution of the difference b/n two sample means.
- Compute mean and standard deviation of the sample distribution
- Compare the values in A and C.

❖ **Central Limit Theorem**

- If we have two normal populations, then sampling distribution of the difference b/n two sample means is normal for all sample size.
- If sample size of each is greater than 30, i.e. n_1 & $n_2 \geq 30$, then sampling distribution of the sample mean is normal.

Normal sampling distribution of the difference b/n two sample means is standardized by using the following formula.

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}}$$

Example: Car stereos of manufacturer A have a mean lifetime of 1400 hours with a standard deviation of 200 hours, while those of manufacturer B have a mean life time of 1200 hours with a standard deviation of 100 hours. If a random sample of 125 stereos of each manufacturer are tested, what is the probability that manufacturer A's stereos will have a mean lifetime which is at least

- 160 hours more than manufacturer B's stereos and
- 250 hours more than the manufacturer B's stereos?

1.2.3. Sampling Distribution of the Sample Proportion (Percentage)

- It is probability distribution of all possible sample proportions of a given (fixed) sample size.
- This distribution is applicable in the analysis of qualitative data, as against quantitative data of analysis.

Example 1

Suppose that we have a population of five students who were asked if they wanted to become managers. The answers were noted as follows

<u>Student</u>	<u>response</u>
A	YES
B	NO
C	YES
D	YES
E	NO

Required:

- A. Find population proportion?
- B. Construct sampling distribution of sampling proportion for sample size of 4?
- C. Compute mean and standard deviation of sampling distribution of sample proportion?
- D. What relation do you observe between population proportion and mean of sampling distribution of sample proportion?

Example 2

Six applicants have applied for a recent vacancy post of Assistant Professor in Business Management department. The department requires only two professors. Their CV for marital status is as follows:.

<i>Applicant</i>	<i>Marital Status</i>
Abebe -----	Single
Belete-----	Married
Chala-----	Single
Debele-----	Married
Elias-----	Married
Feyisa-----	Married

Now a days, psychologists and job analysts recommend the married employees. So, we are interested in the number of married applicants.

Required

- a) Compute population proportion
- b) Construct sampling de-at-ruction of sample proportion
- c) Find mean and standard error of the sampling distribution of sample proportion?

❖ Central Limit Theorem

Central limit theorem is also applicable for sampling distribution of the sample proportion.

1. If the population is normal distribution, then sampling distribution of the sample proportion (distribution of the sample proportion) is also normal for all sample sizes.
2. The sampling distribution of sample proportion is practically normal if the sample size $n \geq 30$.

Normal sampling distribution of the sample proportion can be converted in to standard sampling distribution of the sample proportion by:

$$Z = \frac{\text{Value of RV} - \text{Mean of RV}}{\text{Standard deviation of the RV}}$$

$$= \frac{\bar{p} - \mu_{\bar{p}}}{\sigma_{\bar{p}}}$$

Exercise

A study conducted among FBE students revealed that 45% of them believed that "Introduction to Business statistics" course should be included in the curriculum. A random sample of 80 such students was taken.

Required

- a) What is the probability that
 - i. More than half of them share this belief?
 - i. Greater than 35% of them share this belief
 - i. Between 55% and 60% of them share this belief
- b) About what two proportions will
 - i. 68% of them share this belief?
 - i. Virtually all of them share this belief?

1.2.4 Sampling Distribution of the Difference b/n two Sample Proportions

Example

An instructor giving tutorial class for female students wants the difference in the analytical and interpretation skill for statistical numbers between BA and accounting students. After a careful analysis the teacher found out the following information concerning their skill.

Business Administration					
Name	A	B	C	D	E
Skill	YES	NO	YES	NO	YES

Accounting					
Name	F	G	H	I	J
Skill	NO	YES	NO	YES	NO

Sampling is to be done without replacement and sample sizes of four students are selected from each department.

Required:

- A. Calculate the difference b/n the population proportions.
- B. Construct sampling distribution of the difference b/n the sample proportion for a sample size of four.
- C. Calculate the mean and standard error of the difference b/n the sample proportion.

Chapter Four

Statistical Estimations

Introduction

In business, there are several situations where managers have to make quick estimates. Since their estimates have an impact on the success and failure of their enterprises, they have to take sufficient care to ensure that their estimates are not far away from the actual outcome. One thing we should know here is that, the estimates are made without complete information (from the samples) and with a great deal of uncertainty about the eventual outcome. Therefore, in this chapter we are going to see different methods of statistical estimations.

Basic definitions

Statistical inference (inferential statistics): is a method that is used to determine something about the population on the basis of sample results drawn from the same population. In statistical inference statements about unknown population parameter is derived from information obtained in a random sample selected from the population.

There are two types of inferential methods

1. Estimation
2. Hypothesis testing

Estimation: It is the process of approximating unknown population parameter based on sample known and calculated sample statistics.

There are two types of estimation: Point estimation and interval estimation.

Point estimation

Exercise 1: *The elements in a random sample are 4 10 11 13 16 18. Compute the following point estimates.*

- a. *The estimate of the population mean*
- b. *The estimate of the population standard deviation*
- c. *The estimate of the standard error of the mean*

Exercise 2: *An attempt to investigate how many of the managers in both private and government institutions in Addis Ababa are management graduates revealed that 10% of the 20 sample managers were*

management graduates. Compute the point estimates of

- a. Population proportion
- b. Standard error of the proportion

Interval estimates

An interval estimates consists of two values between which the true population value lies with some stated level of significance.

Interval estimation of the population mean

- An interval estimate of the population mean (μ) is an interval values **a** to **b** with in which an unknown population mean is expected to lie.
- The interval is an inference based on
 - i. The value of the sample mean ($\bar{x}$) of a simple random sample selected from the population, and
 - ii. Known fact about the sampling distribution of the sample mean.
- We cannot be 100% certain that such an interval is correct, because the interval is estimated by using sample results and the samples are some parts or portions of a population. A confident interval estimate of the population mean is an interval estimate together with a statement how confident (e.g. 90%, 95%) we are that the interval is correct.
- The choice of methods in constructing a confidence interval for the population mean(μ) depends up on
 - ✓ Whether or not the population is normal and
 - ✓ Whether the population standard deviation(δ_x) is known or unknown

Case 1: Large sample case ($n \geq 30$), with Standard deviation (δ_x) known

$$CI : \mu = \bar{x} \pm z_{a/2} \frac{\sigma}{\sqrt{n}}$$

$\bar{x}$ = the sample mean

c = the confidence coefficient

$z_{a/2}$ = the z value providing an area of $a/2$ in the upper tail of the standard normal probability distribution

δ = the population standard deviation

n = the sample size

Example 1

Ethiopian management institute (EMI) wishes to have information on the mean income of managers in small scale industries. A random samples of 256 managers revealed a sample mean of birr 45,420. The population standard deviation is 2050.

Required: develop a confidence interval estimate of the mean income of managers in small scale industries with

- i. 95% confidence coefficient
- ii. 90% confidence coefficient
- iii. $\alpha=0.05$

Exercise: A time study has been carried out at a manufacturing plant to determine how long it takes on average to assemble a tractor. Random sample (time in hours) were

3.6 4.2 4 3.5 3.8 3.1

Suppose the population of assembly times has a normal distribution whose standard deviation is 0.30 hour.

Required: Construct a confidence interval estimate for

- a. $C=0.98$
- b. $\alpha=0.1096$
- c. 68% of confidence

Case 2: Large sample case ($n \geq 30$), with Standard deviation (δ_x) unknown

In most applications the value of the population standard deviation is unknown. We simply use the value of the sample standard deviation, s , as the point estimate of the population standard deviation.

$$CI: \mu = \bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

Example: National Discount has 260 retail outlets throughout the United States. National evaluates each potential location for a new retail outlet in part on the mean annual income of the individuals in the marketing area of the new location. Sampling can be used to develop an interval estimate of the mean annual income for individuals in a potential marketing area for National Discount. A sample of size $n = 36$ was taken. The sample mean is \$21,100 and the sample standard deviation(s) is \$4,500. Develop a 95% confidence interval estimate for the mean annual income of individuals.